Survey of undergraduate teaching (PTAS)

Start of Block: Information sheet 1

Q1 Survey of Undergraduate Teaching of Quantitative Study Design and Data Analysis for Life Sciences across the UK

Study goals: We aim to determine the overall level of training in quantitative **Study Design and Data Analysis**1 (defined below) across undergraduate degrees for four disciplines of the life sciences.

Study Design: the structure of a quantitative study that allows it to answer a scientific question/test a hypothesis; e.g., appropriate selection of experimental methods, control groups, methods of sampling or obtaining subjects, how to assign subjects to treatment groups, power analysis.

Data Analysis: the process of analysing the data obtained by a quantitative study. E.g., choice of appropriate statistical tests and unbiased interpretation of results.

We are surveying staff (not students) to characterise teaching in Study Design and Data Analysis. Both the topics covered and general approaches to teaching will be documented in our survey. **The data gathered here will reveal cross-discipline perspectives to improve undergraduate training in quantitative Study Design and Data Analysis for life sciences across the UK**, and will benefit research reproducibility, student experience, learning outcomes and employability.

The data collected will **not** be used to rank universities and **all reported results will be anonymized**. Please click *Next* for more detail on how the data will be used.

1. We broadly define a "study" as a quantitative approach to answer a question, e.g., a manipulative experiment, observational study, longitudinal analysis, etc.

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Q2 Survey of Undergraduate Teaching of Study Design and Data Analysis for Life Sciences across the UK

Study scope

We are studying **undergraduate teaching** of quantitative Study Design and Data Analysis in four disciplines (and sub-disciplines) across every university in the United Kingdom. The four disciplines are Psychology, Biomedical Sciences, Biological Sciences, and Medicine.

How will the results be used?

To develop Life Sciences across the UK, we will (1) publish our results in appropriate **peer-reviewed articles** and (2) disseminate our findings on teaching Study Design and Data Analysis through **cross-disciplinary, UK-wide meetings**. The meetings will facilitate discussion to improve teaching in this area across Psychology, Biology, Biomedical Sciences and Medicine by sharing resources and experience in light of our study's findings.

How you can help

We greatly appreciate your time and effort in completing this survey. We hope this project will provide tools for community development, from and for the community of life scientists across the UK. In this light, **we would like to invite anyone interested in contributing to this study (beyond completing the survey) to join our group**. If you are interested in contributing, please contact us at fmcmanu2@ed.ac.uk and tell us how you might be able to help!

How we can help you

Please contact Franziska at fmcmanu2@ed.ac.uk if you encounter any difficulty with completing this survey. We are happy to arrange an online meeting with you to discuss the survey, and we can complete the survey on your behalf during a conversation with you.

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Q73 We expect the survey will take **about 20-30 minutes** to complete. Where possible, we ask respondents to work with colleagues to complete the survey when one respondent lacks complete knowledge of teaching in quantitative study design and analysis over a student's whole degree within your (sub)discipline.

Please Note: *You may leave and return to the survey as many times as you like.* ***Your progress will be saved for two weeks since the last edit.***

Page Break

Q80 **Please** click here **to download the consent form.**

Q3 I have read the consent form and consent to taking part in this study.

- ○ Yes (1)
- ○ No (2)

Skip To: End of Survey If I have read the consent form and consent to taking part in this study. = No

End of Block: Information sheet 1

Start of Block: Block 1: Personal information

Q65 **1. Personal information**

We collect personal information only to help us organize the data and follow up with respondents, if necessary. All reported results (e.g., published or presented in a talk) will be anonymized and all stored personal data will be deleted one year after publication (as per the consent form).

Q4 a) Your University:

_____

Q5 b) Your Name:

_____

Q6 c) Your Email:

_____

Q7 d) Role connecting you to the teaching of **quantitative Study Design and Data Analysis** (e.g., lecturer, course organiser, director of teaching):

_____

Q8 e) How many years have you taught / organised / overseen teaching in **quantitative Study Design and Data Analysis**?

_____

End of Block: Block 1: Personal information

Start of Block: Block 2: Discipline

Q66 **2. Disciplines in the Life Sciences**

Q9 a) For which **Discipline** are you completing this questionnaire?
*If you teach in more than one discipline then **please indicate for which discipline the answers for the current survey apply**.*

- ○ Psychology (1)
- ○ Biomedical Sciences (e.g., Neuroscience, Reproductive Biology, Pharmacology, Infectious Diseases) (2)
- ○ Biological Sciences (e.g., Ecology, Evolution, Plant Sciences, Molecular Biology). (3)
- ○ Medicine (4)

Q10 Discipline definitions
Psychology: Undergraduate degree in psychology that is accredited by the British Psychological Society (BPS).
Biomedical Sciences: The study of biology with a long-term focus to improve human and animal health by improving understanding that benefits healthcare (e.g., Medical Science) or public health (e.g., Neuroscience, Reproductive Biology, Pharmacology, Infectious Diseases).
Biological Sciences: Biology, broadly defined, is the study of life. However, for the purposes of our survey, we distinguish “biology” from “biomedical sciences”. Hence, we restrict our

definition of Biological Sciences to the study of biology without a long-term focus of improving human or animal health (e.g., Ecology, Evolution, Plant Sciences, Molecular Biology, and Environmental Science when taught with a biological focus).
Medicine: a medical degree accredited by the General Medical Council.

Q11 b) We recognise that at some institutions, students in one discipline receive identical training to students in another discipline. Please indicate here if this is the case and list the disciplines to which this applies.

Page Break

Display This Question:

If a) For which Discipline are you completing this questionnaire? If you teach in more than one disc... = Psychology

Q12 **Psychology:** Undergraduate degree in psychology that is accredited by the British Psychological Society (BPS).

Display This Question:

If a) For which Discipline are you completing this questionnaire? If you teach in more than one disc... = Biomedical Sciences (e.g., Neuroscience, Reproductive Biology, Pharmacology, Infectious Diseases)

Q13 **Biomedical Sciences:** The study of biology with a long-term focus to improve human and animal health by improving understanding that benefits healthcare (e.g., Medical Science) or public health (e.g., Neuroscience, Reproductive Biology, Pharmacology, Infectious Diseases).

Display This Question:

If a) For which Discipline are you completing this questionnaire? If you teach in more than one disc... = Biological Sciences (e.g., Ecology, Evolution, Plant Sciences, Molecular Biology).

Q14 **Biological Sciences:** Biology, broadly defined, is the study of life. However, for the purposes of our survey, we distinguish “biology” from “biomedical sciences”. Hence, we restrict our definition of Biological Sciences to the study of biology without a long-term focus of improving human or animal health (e.g., Ecology, Evolution, Plant Sciences, Molecular Biology, and Environmental Science when taught with a biological focus).

Display This Question:

If a) For which Discipline are you completing this questionnaire? If you teach in more than one disc... = Medicine

Q15 **Medicine**: a medical degree accredited by the General Medical Council.

Q16 c) This is the definition you have chosen. Please indicate whether this is correct.

- ○ Yes, this is correct. (1)
- ○ No, I would like to choose again. (2)

Display This Question:

If a) For which Discipline are you completing this questionnaire? If you teach in more than one disc... = Biological Sciences (e.g., Ecology, Evolution, Plant Sciences, Molecular Biology).

Q72 d) Does the student cohort for which you are answering this survey learn about **ecology and/or evolution**?

- Yes - ecology (1)
- Yes - evolution (2)
- Yes - both (3)
- No - neither (4)

Q92 *Please note: you will not be able to go backwards in the survey after this point. If there is an answer on a previous page that you would like to change, then please do so now.*

End of Block: Block 2: Discipline

Start of Block: Block 3: Subdisciplines

Q17 3. Do the majority of students within your discipline receive the same compulsory ("core") training in quantitative study design and analysis?

For example, if students in a Biological Sciences programme receive different compulsory Study Design/Data Analysis teaching depending on whether they specialise in Zoology or Plant Sciences, then "No" would be selected.

If you are unsure, then please select "No" for more information.

- Yes (1)
- No (2)

Display This Question:

If 3. Do the majority of students within your discipline receive the same compulsory ("core") traini... = No

Q18 Sub-disciplines (or Specializations)- IMPORTANT:

This survey documents teaching for **(largely) independent cohorts** of undergraduate students. In other words, our analysis uses 'student cohort' as a unit of observation. We recognize, however, that sometimes **multiple student cohorts** receive (different) training independently within a discipline at a university. We therefore require survey respondents to indicate the sub-discipline (i.e., programme/specialisation) to which the survey answers apply, when training in quantitative study design and analysis differs among sub-disciplines within the selected discipline at your institution. Consequently, we may ask for the survey to be completed multiple times (once per sub-discipline) for a given discipline at a University. **In such cases, we are happy for you to complete separate instances of this survey for various sub-disciplines at your university**. Alternatively, please let us know which staff you feel are most qualified to complete it for a given student cohort/sub-discipline. In both cases, we will send a new link for each sub-discipline.

If you are unsure, please click here for two examples.

Display This Question:

If 3. Do the majority of students within your discipline receive the same compulsory ("core") traini... = No

Q19 3b) If applicable, please specify a sub-discipline. Otherwise, please write "NA".

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Display This Question:

If 3. Do the majority of students within your discipline receive the same compulsory ("core") traini... = No

Q20 3c) If you feel the two 'sub-discipline' examples provided **do not apply** to your Discipline at your university, then please describe the situation at your university.

_____

End of Block: Block 3: Subdisciplines

Start of Block: Block 4: Teaching questions

Q21 **4. Teaching**

Unless noted specifically, all questions refer to compulsory ('core') teaching in your Discipline. Please pay attention as to whether the question is asking about Study Design or Data Analysis instruction (or both).

Study Design: *the structure of a quantitative study that allows it to answer a scientific question/test a hypothesis; e.g., appropriate selection of experimental methods, control groups, methods of sampling or obtaining subjects, how to assign subjects to treatment groups, power analysis.*

Data Analysis: *the process of analysing the data obtained by a quantitative study. E.g., choice of appropriate statistical tests and unbiased interpretation of results.*

Page Break

Q22 a) i. At your institution, are undergraduates in your Discipline (Sub-Discipline) taught **Study Design** at any point in their degree?

- ○ Yes (1)
- ○ No (2)

Q23 *If applicable:*
a) ii. Is the **Study Design** content assessed?
Please select all that apply.

- ☐ Yes, with assessment dedicated to this topic (1)
- ☐ Yes, as part of a broader assessment (e.g. a few questions in an exam; as marked component of a practical / lab report, etc) (2)
- ☐ No (3)

Page Break

Q24 b) i. At your institution, are undergraduates in your Discipline (Sub-Discipline) taught **Data Analysis** at any point in their degree?

- ○ Yes (1)
- ○ No (2)

Q25 *If applicable:*
b) ii. Is the **Data Analysis** content assessed?
Please select all that apply.

- ☐ Yes, with assessment dedicated to this topic (1)
- ☐ Yes, as part of a broader assessment (e.g. a few questions in an exam; as marked component of a practical / lab report, etc) (2)
- ☐ No (3)

Display This Question:

If a) i. At your institution, are undergraduates in your Discipline (Sub-Discipline) taught Study De... = Yes

Or b) i. At your institution, are undergraduates in your Discipline (Sub-Discipline) taught Data Ana... = Yes

Q26 c) How does the degree package its instruction in **Data Analysis** and/or **quantitative Study Design**?

- ○ Provides one or more teaching unit devoted to these topics (1)
- ○ Instruction in Data Analysis / quantitative Study Design is embedded in teaching unit(s) that encompass additional topics (2)
- ○ Both (please provide details below) (3)

Display This Question:

If b) i. At your institution, are undergraduates in your Discipline (Sub-Discipline) taught Data Ana... = Yes

Q27 d) How would you describe the dominant approach to teaching **Data Analysis**?

- ○ Mathematical (e.g. teaching the mathematics of the Normal Distribution, or proofs of a t-test) (1)
- ○ Functional (e.g. emphasising how experimental design matches with an appropriate analysis approach, without emphasising the mathematics behind the analysis) (2)
- ○ An equal balance of Mathematical and Functional. (3)
- ○ Both, emphasising Mathematical (4)
- ○ Both, emphasising Functional (5)

Q28 e) In addition to core teaching, do students have the opportunity to receive more instruction in **Data Analysis** and/or quantitative **Study Design** through elective courses?

- ○ Yes (1)
- ○ No (2)

Q81 [optional] Is there anything else that you would like to add?

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End of Block: Block 4: Teaching questions

Start of Block: Block 5: Amount of teaching

Q67 **5. Amount of teaching**

Q29 a) How many years is the **undergraduate degree** for which you are answering this survey?
Please disregard optional placement years/years abroad. For an integrated masters, please answer for the undergraduate section of the course only.

▼ 3 (3) ... Other (9)

Q82 [optional] If you selected Other, then please provide more detail here.

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Page Break

Q85 b) i) How many credits of instruction in **quantitative Study Design** do students receive in **core (compulsory) teaching** across the entire undergraduate degree?

As a guideline: 1 credit = approximately 10 hours of study, including independent study.
A course that runs for one semester is usually 20 credits (200 hours of study time).
A full semester of teaching often consists of 60 credits.

*Please give an **approximate** number of credits. We are aware that you may not know details about the whole degree. If possible, please confer with a colleague to help answer this question.*

▼ Less than 10 (1) ... 80+ (11)

Q89 b) ii) What is the format of undergraduate instruction in **quantitative Study Design** that students receive in core (compulsory) teaching over their entire degree?
Please select all that apply.

- ☐ Lectures (1)
- ☐ Practicals/workshops/tutorials (2)
- ☐ Additional (e.g. students watch videos for 'flipped classroom') (3)

Page Break

Q86 c) i) How many credits of undergraduate instruction in **Data Analysis** do students receive in **core (compulsory) teaching** across the entire undergraduate degree?

As a guideline: 1 credit = approximately 10 hours of study, including independent study.
A course that runs for one semester is usually 20 credits (200 hours of study time).
A full semester of teaching often consists of 60 credits.

*Please give an **approximate** number of credits. We are aware that you may not know details about the whole degree. If possible, please confer with a colleague to help answer this question.*

▼ Less than 10 (1) ... 80+ (11)

Q88 c) ii) What is the format of undergraduate instruction in **Data Analysis** that students receive in core (compulsory) teaching over their entire degree?
Please select all that apply.

- ☐ Lectures (1)
- ☐ Practicals/workshops/tutorials (2)
- ☐ Additional (e.g. students watch videos for 'flipped classroom') (3)

Page Break

Q32 d) In which years do students receive **core** instruction in either quantitative **Study Design** or **Data Analysis**?
Please select all that apply.

- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ NA - No instruction provided (5)

Display This Question:

If a) i. At your institution, are undergraduates in your Discipline (Sub-Discipline) taught Study De... = Yes

Q33 e) To your knowledge, has the extent of training in **quantitative Study Design** increased or decreased notably in the last decade?
e.g. Yes - net increase

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Display This Question:

If b) i. At your institution, are undergraduates in your Discipline (Sub-Discipline) taught Data Ana... = Yes

Q34 f) To your knowledge, has the extent of training in **Data Analysis** increased or decreased notably in the last decade?

e.g. Yes - net increase

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End of Block: Block 5: Amount of teaching

Start of Block: Block 6: Statistical software

Q68 **6. Statistical Software**
Please answer based on teaching across the whole degree.

Q35 a) What format of **statistical software** do students learn to use?

- ○ Script-based (e.g., R, Python, SAS, SPSS using syntax, other) (1)
- ○ Push-button based (e.g., Minitab, JMP, SPSS using the drop-down menus, Excel, Prism, Jamovi, JASP, other) (2)
- ○ Both (3)
- ○ Neither (4)

Q36 *If applicable:*
b) Please indicate the dominant software used over a student's degree.

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End of Block: Block 6: Statistical software

Start of Block: Block 7: Additional opportunities

Q69 **7. Additional Opportunities**

Q37 a) i. Do students have the opportunity to participate in research project(s) pre-final-year (either assisting others with research at their school or conducting their own research project) for which they earn credit towards their degree, where they can apply their **quantitative** Study Design and Data Analysis knowledge?

- Yes (1)
- No (2)

Q90 *If you answered* No*, then please skip to the next page.*

Q38 a) ii. Are these pre-final-year research project(s) activities compulsory?

- Yes (1)
- No (2)

Q39 a) iii. Are they assessed?

- Yes (1)
- No (2)

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Q40 b) i. Do students have the opportunity to conduct an independent final-year research project, where they can apply their Study Design and Data Analysis knowledge?

- ○ Yes - quantitative project (1)
- ○ Yes - quantitative and/or qualitative project (2)
- ○ No (3)

Q91 *If you answered* No*, then please skip to the next page.*

Q41 b) ii. *(if Yes)* Is this final-year independent research project **compulsory**?

- ○ Yes (1)
- ○ No (2)

Q42 b) iii. *(if Yes)* Is this final-year independent research project **assessed**?

- ○ Yes (1)
- ○ No (2)

End of Block: Block 7: Additional opportunities

Start of Block: Block 8 - Teaching content

Q43 **8. Teaching Checklist**

The following questions are based on content across the **entire undergraduate degree** for your discipline (or sub-discipline) at your institution. We recognise that often knowledge about a degree is spread across multiple people, so if you cannot answer a question yourself, then please try to contact a colleague who does know.

Reminder: You may leave and return to the survey as many times as you like. Your progress

will be saved for two weeks since the last edit.

Page Break

Q74 Content:
a) Study Design
b) Interpreting Results
c) Data Handling
d) Statistical Concepts
e) Statistical Tests and Approaches
f) Questionable Research Habits
g) Open Research
h) Ethics

Q44 **a) Study Design.** Do students learn about each of the following topics during their undergraduate years?
Please only select Don't Know *if you truly cannot find an answer (e.g by asking a colleague).*

Observational vs. experimental approaches (1)	○ Yes (1)	○ No (2)	○ Don't Know (3)
How to match adequate statistical methods to Study Design and to the scientific question / hypothesis (2)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Select appropriate control groups (3)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Need for replication in a study (4)	○ Yes (1)	○ No (2)	○ Don't Know (3)
How to select study materials (e.g., subjects, cell lines) that are representative of study's goals (5)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Identifying confounding variables, or use methods to avoid confounding variables in experimental design (6)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Identifying and accounting for sources of non-independence (pseudo-replication) (7)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Power analysis: consequences of low-powered studies (8)	○ Yes (1)	○ No (2)	○ Don't Know (3)

Power analysis: application (9)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Blinding (10)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Use of randomization throughout an experiment (e.g., random allocation of subjects to treatment, randomize location of subjects in experimental area, etc) (11)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Blocking (12)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Balanced vs. unbalanced designs (13)	○ Yes (1)	○ No (2)	○ Don't Know (3)

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Q76 Content:
a) Study Design
b) Interpreting Results
c) Data Handling
d) Statistical Concepts
e) Statistical Tests and Approaches
f) Questionable Research Habits
g) Open Research
h) Ethics

Q45 **b) Interpreting Results.** Do students learn about each of the following topics during their undergraduate years?
Please only select Don't Know *if you truly cannot find an answer (e.g by asking a colleague).*

Interpreting results in light of p-values (1)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Interpreting results in light of effect size with confidence intervals (2)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Basing conclusions on multiple lines of evidence; e.g., effect size vs. p-value; complementary experiments (3)	○ Yes (1)	○ No (2)	○ Don't Know (3)

Q46 **c) Data Handling.** Do students learn about each of the following topics during their undergraduate years?
Please only select Don’t Know *if you truly cannot find an answer (e.g by asking a colleague).*

Data cleaning and preparation (1)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Visualising data (2)	○ Yes (1)	○ No (2)	○ Don't Know (3)

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Q78 Content:
a) Study Design
b) Interpreting Results
c) Data Handling
d) Statistical Concepts
e) Statistical Tests and Approaches
f) Questionable Research Habits
g) Open Research
h) Ethics

Q47 **d) Statistical Concepts.** Do students learn about each of the following topics during their undergraduate years?
Please only select Don't Know *if you truly cannot find an answer (e.g by asking a colleague).*

Exploratory vs. confirmatory studies (1)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Descriptive statistics (2)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Probability (3)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Distributions (4)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Central limit theorem (5)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Estimating with uncertainty (understanding standard error and confidence intervals) (6)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Null hypothesis testing vs. parameter estimation (7)	○ Yes (1)	○ No (2)	○ Don't Know (3)

Q48 **e) Statistical Tests and Approaches.** Which of the following topics do students learn during their undergraduate years?
Please only select Don't Know *if you truly cannot find an answer (e.g by asking a colleague).*

t-tests (1)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Frequency tests (2)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Correlation (3)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Simple Linear Regression (General Linear Model with 1 covariate) (4)	○ Yes (1)	○ No (2)	○ Don't Know (3)
ANOVA (1-Factor General Linear Models) (5)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Data transformation (6)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Non-parametric statistics (7)	○ Yes (1)	○ No (2)	○ Don't Know (3)
General linear models (i.e., Gaussian data) with multiple independent variables (8)	○ Yes (1)	○ No (2)	○ Don't Know (3)
General linear mixed effects models (9)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Non-linear regression (10)	○ Yes (1)	○ No (2)	○ Don't Know (3)

Generalized linear (mixed) models (11)	Yes (1)	No (2)	Don't Know (3)
Generalized estimating equations (12)	Yes (1)	No (2)	Don't Know (3)
Path analysis (13)	Yes (1)	No (2)	Don't Know (3)
Structural equation modelling (SEM) (14)	Yes (1)	No (2)	Don't Know (3)
Principal component analysis (PCA) (15)	Yes (1)	No (2)	Don't Know (3)
Exploratory factor analysis (EFA) (16)	Yes (1)	No (2)	Don't Know (3)
Confirmatory factor analysis (CFA) (17)	Yes (1)	No (2)	Don't Know (3)
Likelihood (18)	Yes (1)	No (2)	Don't Know (3)
Bayesian statistics (19)	Yes (1)	No (2)	Don't Know (3)
Computational methods (e.g., bootstrapping, randomization tests) (20)	Yes (1)	No (2)	Don't Know (3)
Meta-analysis (21)	Yes (1)	No (2)	Don't Know (3)
Machine Learning (22)	Yes (1)	No (2)	Don't Know (3)

Deep learning (23)	○ Yes (1)	○ No (2)	○ Don't Know (3)

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Q79 Content:
a) Study Design
b) Interpreting Results
c) Data Handling
d) Statistical Concepts
e) Statistical Tests and Approaches
f) Questionable Research Habits
g) Open Research
h) Ethics

Q49 **f) Questionable Research Habits.** Do students learn about each of the following topics during their undergraduate years?
Please only select Don't Know *if you truly cannot find an answer (e.g by asking a colleague).*

Hypothesising after results known ('HARKing') (1)	○ Yes (1)	○ No (2)	○ Don't Know (3)
p-hacking (2)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Problems arising from unreported variables (3)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Problems arising from unreported covariates (4)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Problems arising from unreported analyses (5)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Rounding p-values (6)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Problems arising from excluding data to obtain a more desirable conclusion (7)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Adding data after analysis to improve p-value (8)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Problems arising from switching analysis to obtain a more desirable conclusion (9)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Undisclosed problems (10)	○ Yes (1)	○ No (2)	○ Don't Know (3)

Data fabrication (11)	○ Yes (1)	○ No (2)	○ Don't Know (3)

Q50 **g) Open Research.** Do students learn about each of the following topics during their undergraduate years?
Please only select Don't Know *if you truly cannot find an answer (e.g by asking a colleague).*

Study pre-registration (1)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Registered reports (2)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Data sharing (3)	○ Yes (1)	○ No (2)	○ Don't Know (3)
"FAIR" data availability practices (4)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Open notebooks (5)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Data organisation (e.g. file and folder structure) (6)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Metadata (e.g. a data dictionary) (7)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Appropriate data storage (8)	○ Yes (1)	○ No (2)	○ Don't Know (3)

Q51 **h) Ethics.** Do students learn about each of the following topics during their undergraduate years?
Please only select Don't Know *if you truly cannot find an answer (e.g by asking a colleague).*

Informed consent and participant rights (1)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Ethical data storage and management (e.g., anonymity, meeting GDPR standards) (2)	○ Yes (1)	○ No (2)	○ Don't Know (3)
Ethical principles (e.g. respect, competence, responsibility, and integrity) (3)	○ Yes (1)	○ No (2)	○ Don't Know (3)

Q52 i) **Other.** Is there anything else that you teach within Study Design/Data Analysis that you think is relevant, but is not listed?

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End of Block: Block 8 - Teaching content

Start of Block: Block 9: Contacts

Q71 **9. Willing to be contacted?**

Q53 a) Are you willing to be contacted for a follow-up discussion?

○ Yes (1)

○ No (2)

Q83 b) Is there another sub-discipline within your discipline that we should be aware of?
For example, Zoology and Plant Sciences would count as different sub-disciplines within the Biological Sciences discipline if they differed in their core (compulsory) Study Design/Data Analysis teaching.

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Q54 c) Can you recommend someone who teaches **quantitative Study Design and Data Analysis** in another discipline at your institution or at another institution, and who might be interested in taking part in the study?

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End of Block: Block 9: Contacts

Start of Block: Block 10: Optional questions

Q55 **10. Optional Questions**
The following questions are optional. Please continue to the next page if you do not wish to answer them.

Q56 a) What challenges have you faced when teaching quantitative Study Design and Data Analysis and what approaches have you used to overcome them (or not)?

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Q57 b) Have you changed your approach to teaching quantitative Study Design and Data Analysis over your career? If so, what have you learned from your change(s)?

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Q58 c) Do you have anything further you wish to share?

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End of Block: Block 10: Optional questions

Start of Block: Block 11: Thank you

Q59

Thank you for participating.

If you are interested in contributing to the project, please contact us at f.mcmanu2@ed.ac.uk to talk about how we may work together!

Please press *Submit* when you are ready to send us your final answers.

End of Block: Block 11: Thank you